

TrackMind AI - Total System Analysis

AI-Powered Automatic Block Planning & Disconnection Management System (BDMS)

Ministry of Railways | Smart India Hackathon PS 26027 | Full Architectural Specification

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1. Executive Overview & Problem Statement

In Indian Railways, three core infrastructure engineering departments constantly require maintenance blocks (temporary corridor closures):

- **Engineering (Track / P-Way / TMS):** Rail fractures, deep screening, track packing, rail renewals, turnouts, and bridge structural repairs.
- **Signal & Telecom (S&T / SMMS):** Electronic interlocking, point machines, track circuits, axle counters, and signal lamp maintenance.
- **Traction Distribution (TRD / TDMS):** Overhead Equipment (OHE) wire wear, catenary dropper alignment, power shutdowns, and insulator washing.

Historically, these departments requested blocks in silos through BDMS with zero real-time cross-departmental coordination. This resulted in scheduling collisions, severe passenger train delays, and unexploited corridor windows. TrackMind AI unifies public field defect reporting, departmental engineering assessments, automated candidate group classification, mathematical CP-SAT optimization, live GIS train telemetry, and multilingual conversational decision support.

2. Comprehensive Technology Stack Architecture

Layer	Technologies Used	Role & Operational Implementation
Frontend UI & Layout	Streamlit (1.38+), Custom CSS3	Reactive full-stack web dashboard with wide 1160px card containers and view routing.
Geospatial GIS Engine	Folium (0.20+), Leaflet.js	Satellite multi-track map, station pill markers (DivIcon), live train vector tracking & HUD.
Operations Research Solver	Google OR-Tools CP-SAT	Constraint Programming engine ensuring zero collisions and passenger timetable protection.
Machine Learning Layer	Scikit-Learn (RandomForest, Isolation)	Failure risk prediction (RandomForest) and track anomaly defect clustering (IsolationForest).
Data Processing & Datasets	Pandas (2.2+), NumPy, Faker	Vectorized DataFrame transformations and 2,000+ realistic synthetic railway records.
GenAI & LLM Engine	Groq SDK (LLaMA-3.3-70B)	Sub-second AI reasoning with automated fallback to LLaMA-3.1-8B, Mixtral, and Gemma2.
Knowledge RAG Engine	Custom Intent Parser + SQL RAG	Live database querying engine against railway.db for zero-hallucination factual stats.
Voice Interaction	Web Speech API (STT / TTS)	Browser-native multilingual speech recognition & synthesis (English, Hindi, Telugu).
Database & Storage Layer	SQLite3 (WAL Mode)	Relational database with Write-Ahead Logging and 30s busy timeout for concurrent safety.
Security & Cryptography	Bcrypt (4.2+)	Salted password encryption and Role-Based Access Control (Admin, Eng, S&T, TRD).
Reporting Engine	ReportLab (4.2+), FPDF2 (2.8+)	Automated server-side generation of downloadable technical specifications and PDF reports.

3. Public Defect Intake & Department Assessment Workflow

The platform strictly partitions reporting observation from engineering severity determination:

- **Public Add Defect Gateway:** Wide 1160px desktop form open to Loco Pilots, Patrol Officers, and public. Restricted strictly to 3 departments (Engineering, S&T, TRD). Zero severity inputs (defaults to Not Yet Assessed).
- **Deterministic Department Routing:** Generates DEF-YYYYMMDD-XXXX and posts a standardized 5-line alert notification to the responsible department queue.
- **AI Advisory Severity Suggestion:** Analyzes structural keywords and safety hazards to provide an advisory severity and safety reasoning with an [Accept AI Suggestion] button without locking the engineer.
- **Department Assessment Authority:** Department engineer conducts technical diagnosis, sets official Severity (Low/Medium/High/Critical), and enters recommended field actions.
- **Block Requisition Creation:** Auto-populates defect observation details, sets duration & track line, and transmits requisition into block_requests_v2 with status SUBMITTED.

4. Candidate Group Preprocessing & Classification Archetypes

When Section Controllers preprocess unprocessed requisitions (app/classification_engine.py), the engine derives safety dependencies and groups them into 4 operational archetypes:

- **ISOLATION:** OHE electrification blocks requiring 25kV power cut, traction isolation, and earthing.
- **PARALLEL (Shadow Blocks):** Co-located multi-department tasks scheduled in unified windows, boosting corridor availability by up to 35%.
- **SEQUENTIAL:** Chained dependent tasks executed in strict chronological order (e.g. Track packing followed by Signal recalibration).
- **REVIEW:** Complex or conflicting requests requiring manual controller intervention or traffic diversion.

All classification states (is_classified, classification_run_id, classified_at) persist in SQLite, ensuring state stability across page refreshes.

5. Mathematical Optimization Engine (Google OR-Tools CP-SAT)

The scheduling core (scripts/optimizer.py & app/block_allocation_engine.py) formulates maintenance planning as a binary integer constraint satisfaction problem (CP-SAT):

- **Hard Constraint 1 (No Collisions):** Each corridor slot holds at most one maintenance block unless merged into an approved shadow block.
- **Hard Constraint 2 (Timetable Protection):** Slots overlapping scheduled passenger train departure windows on the same section are purged via filter_slots_against_timetable().
- **Hard Constraint 3 (Section & Duration Feasibility):** Tasks are scheduled only into slots on their identical section ID with slot duration $\geq$ estimated task duration.
- **Objective Function:** Maximize cumulative priority: Maximize $\sum(x_{ij} * [0.60 * \text{PriorityScore}_i + 0.40 * \text{RiskScore}_i])$.

6. Predictive ML & 32 Autonomous Domain Agents

Every maintenance defect across TMS, SMMS, and TDMS undergoes predictive risk analysis and autonomous agent supervision:

- **RandomForest Risk Score:** Predicts in-service failure probability (0-100%) based on severity, overdue days, and traffic density.
- **IsolationForest Anomaly Detector:** Flags track sections exhibiting statistically abnormal defect clustering.
- **32 Domain Agents Network:** Specialized autonomous Python agents organized into Core Operations, Safety & Compliance, Locopilot Speed & TSR, Resource Logistics, and Passenger Advisory bulletins.

7. Multilingual ChatMind AI & Intent Parsing RAG

Embedded floating conversational assistant in app/chatbot.py with intent parsing and live SQL generation:

- **Multi-Intent Parser:** Extracts entities for TRAIN_SINGLE_QUERY, TRAIN_DELAYED_FILTER_QUERY, DB_QUERY, REQUEST_LOOKUP, and DEFECT_LOOKUP.
- **Live SQL Query Engine:** Executes parameterized queries against railway.db for verified, real-time factual counts and rankings.
- **Voice Interaction:** Browser Web Speech API supporting English (en-IN), Hindi (hi-IN), and Telugu (te-IN).

8. System-Wide Verification & Business Impact

- **18 Automated Test Suites:** 241 test cases covering GIS maps, live telemetry, classification, CP-SAT allocation, notifications, chatbot, and defect ingestion passing at 100%.
- **100% Conflict Elimination:** Zero departmental clashes through mathematical constraint programming.
- **Corridor Capacity Boost:** Shadow block clustering increases effective corridor availability by up to 35%.